Supplementary Information (SI) for Energy Advances.
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Optical constants manipulation of formamidinium lead iodide perovskites: Ellipsometric and spectroscopic twigging

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S1. Fitting of Spectroscopic Ellipsometer data

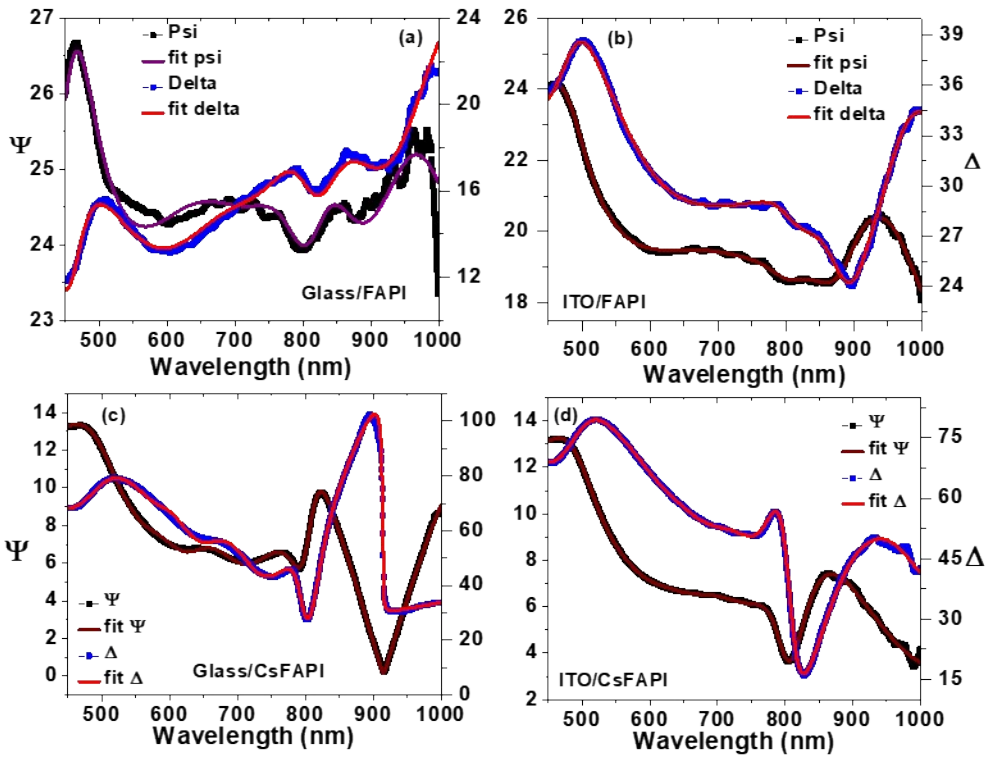

Figure S1. The measured amplitude ratio (ψ) and phase difference (Δ) of the FAPI and CsFAPI thin films on glass and ITO.

Table S1. Tauc-Lorentz fit parameters for FAPI and CsFAPI thin films on glass and ITO.

Sample	Glass/FAPI	ITO/FAPI	Glass/CsFAPI	ITO/ CsFAPI
r (nm)	15	23	21	15
Void (%)	20	23	12	9
ε_∞	1.25	1.24	1.10	1.01
A_1	3.43	30.55	24.53	6.16
E_1	1.55	1.45	1.57	1.58
C_1	0.19	0.44	0.13	0.13
A_2	44.45	52.101	7.60	6.52
E_2	2.18	2.18	2.46	2.45
C_2	4.35	3.70	0.49	0.54
A_3	0.83	1.00	6.50	8.26
E_3	2.70	2.45	3.31	3.18
C_3	0.41	0.49	3.89	0.61
χ^2	0.18	0.03	0.10	0.03

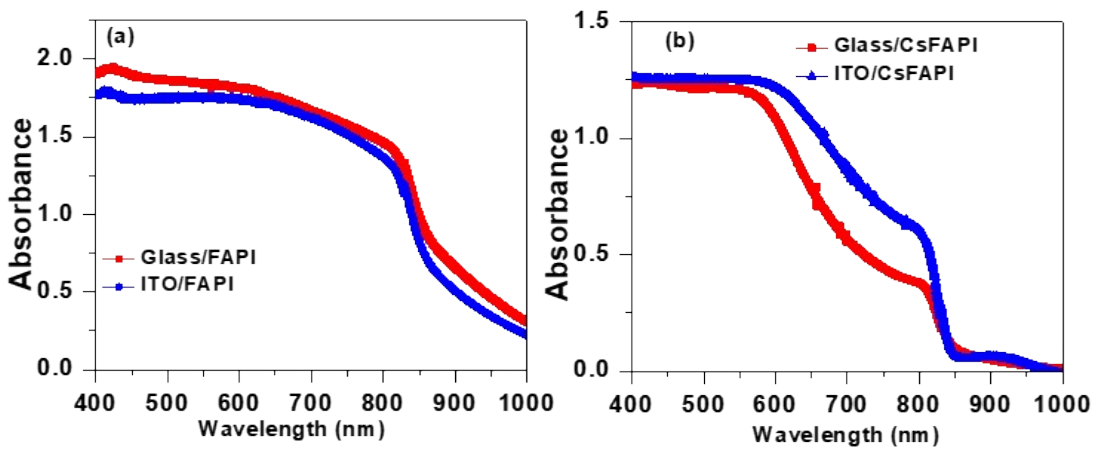

Figure S2. The absorption spectra of (a) $FAPbI_3$, and (b) $CsFAPbI_3$ thin films.

Table S2: Energy bandgap of FAPI and CsFAPI thin films on glass and ITO.

Sample	E_g
Glass/ FAPI	1.49 Ev
Glass/ITO/FAPI	1.50 eV
Glass/CsFAPI	1.59 eV
Glass/ITO/CsFAPI	1.58 eV

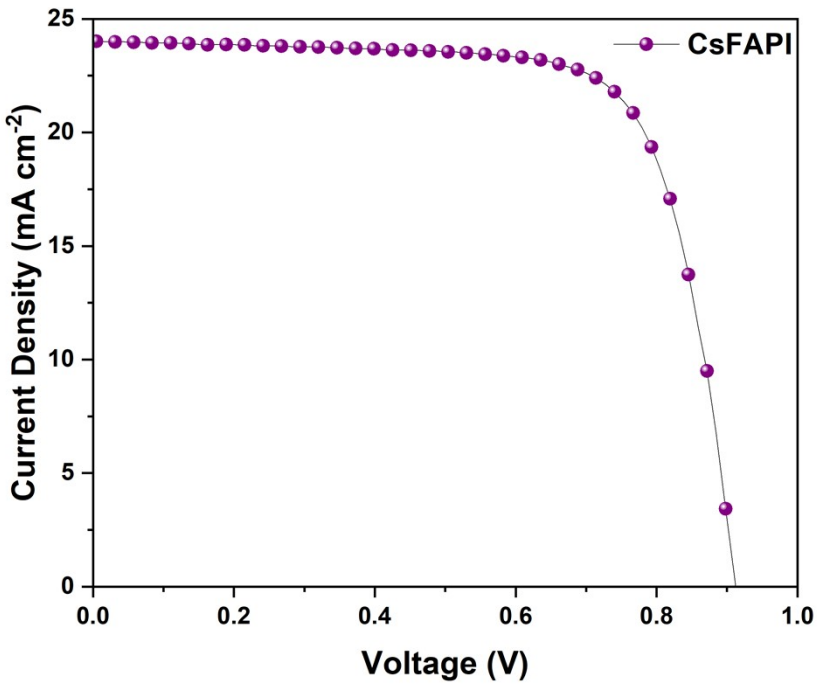

Figure S3. *J-V* characteristics under illumination of a typical CsFAPI-based perovskites solar cell.

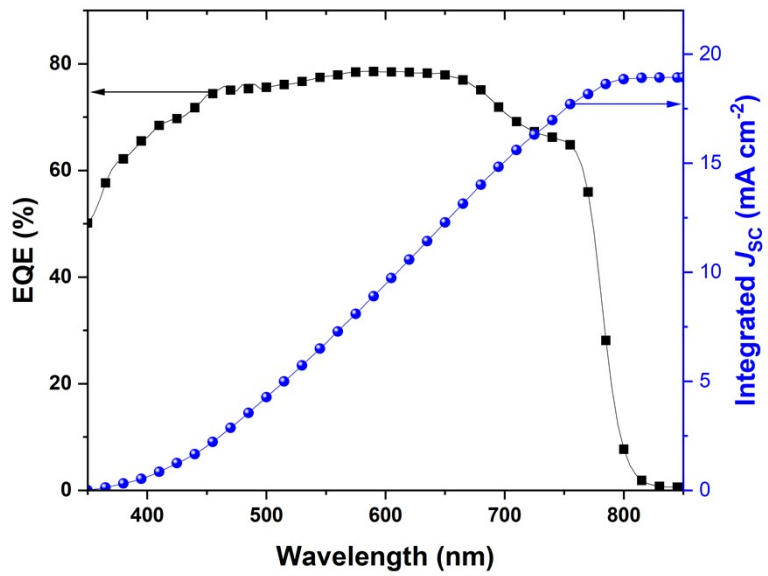

Figure S4. EQE characteristics of a typical CsFAPI-based perovskites solar cell.